

ODE Notes

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Seperable D.E.

$$\frac{dy}{dx} = f(x)g(y)$$

Linear D.E.

$$\sum_{i=1}^n a_i(x)y^i(x) = b(x)$$

First order

$$y'(x) + P(x)y = Q(x)$$

Solved by

$$y(x) = \frac{1}{\mu(x)} \int \mu(x)Q(x)dx$$

where $\mu(x) = e^{\int P(x)dx}$

Bernoulli D.E.

$$y'(x) = P(x)y(x) + Q(x)(y(x))^n$$

For $n = 0, 1$ this is seperable
 $y = 0$ is always a solution

$$\frac{y'(x)}{(y(x))^n} = \frac{p(x)}{(y(x))^{n-1}} + Q(x)$$

Let $v(x) = \frac{1}{(y(x))^{n-1}}$. Then $v'(x) = \frac{(1-n)y'(x)}{(y(x))^n}$
So we get

$$\frac{v'(x)}{1-n} = P(x)v(x) + Q(x)$$

Which is a Linear D.E. in $v(x)$ (solve then sub back to $y(x)$)

Homogeneous Functions

A function $f(x, y)$ is said to be homogeneous of degree n if $f(\lambda x, \lambda y) = \lambda^n f(x, y)$ for all $\lambda \in \mathbb{R}$

Homogeneous D.E.

$$\frac{dy}{dx} = \frac{f(x, y)}{g(x, y)}$$

Where f, g are homogeneous of the same degree

Thus

$$\frac{dy}{dx} = \frac{f(\lambda x, \lambda y)}{g(\lambda x, \lambda y)}$$

Letting $\lambda = \frac{1}{x}$, we make $\frac{dy}{dx} = F(\frac{y}{x})$, a function of $\frac{y}{x}$

Let $v = \frac{y}{x}$ then $v + x \frac{dv}{dx} = \frac{dy}{dx} = F(v)$

Which is separable

$$\frac{dv}{F(v) - v} = \frac{dx}{x}$$